Задание 3

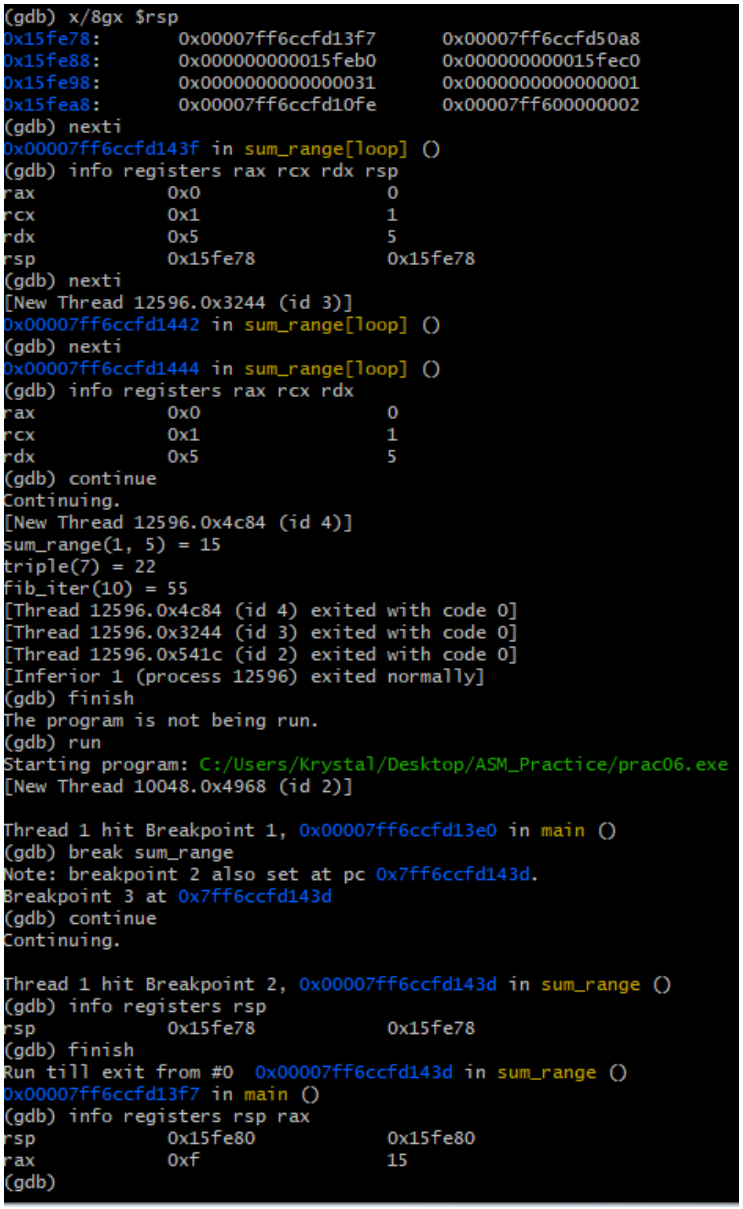
```
Krystal@DESKTOP-TEU6EL3 UCRT64 /c/Users/Krystal/Desktop/ASM_Practice
$ gdb ./prac06.exe
GNU gdb (GDB) 18.1
Copyright (C) 2026 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-w64-mingw32".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.

+------------------------------------------------------------------------------+
| Find the GDB manual online at:                                               |
| http://www.gnu.org/software/gdb/documentation/.                              |
| For help, type "help".                                                       |
| Type "apropos word" to search for commands related to "word".                |
+------------------------------------------------------------------------------+

Reading symbols from ./prac06.exe...
(gdb) break main
Breakpoint 1 at 0x1400013e0
(gdb) run
Starting program: C:/Users/Krystal/Desktop/ASM_Practice/prac06.exe
[New Thread 12596.0x541c (id 2)]

Thread 1 hit Breakpoint 1, 0x00007ff6ccfd13e0 in main ()
(gdb) info registers
rax            0x7ff6ccfd5118      140697977835800
rbx            0x7ff6ccfd50a8      140697977835688
rcx            0x1                 1
rdx            0x5da950            6138192
rsi            0x5df530            6157616
rdi            0x31                49
rbp            0x1                 0x1
rsp            0x15fea8            0x15fea8
r8             0x5df530            6157616
r9             0x5dea50            6154832
r10            0x8310adc476c4      144107658049220
r11            0x7ffd4f4cdf27      140725933891367
r12            0x5da8d0            6138064
r13            0x1                 1
r14            0x7ff6ccfd50a0      140697977835680
r15            0x1                 1
rip            0x7ff6ccfd13e0      0x7ff6ccfd13e0 <main>
eflags         0x202               [ IF ]
cs             0x33                51
ss             0x2b                43
ds             0x2b                43
es             0x2b                43
fs             0x53                83
gs             0x2b                43
(gdb) break sum_range
Breakpoint 2 at 0x7ff6ccfd143d
(gdb) continue
Continuing.

Thread 1 hit Breakpoint 2, 0x00007ff6ccfd143d in sum_range ()
(gdb) info registers rcx rdx rsp rbp
rcx            0x1                 1
rdx            0x5                 5
rsp            0x15fe78            0x15fe78
rbp            0x15fea0            0x15fea0
```



Задание 5

Параметр	fib(4)	fib(3)	fib(2)	fib(1)
RBP	0x15fe70	0x15fe30	0x15fdf0	0x15fdb0
RSP	0x15fe40	0x15fe00	0x15fdc0	0x15fd80
Адрес возврата	0x00007ff78cf213f2	0x00007ff78cf21420	0x00007ff78cf21420	0x00007ff78cf21420

```
Reading symbols from ./fib.exe...
(gdb) break fib
Breakpoint 1 at 0x140001408
(gdb) run
Starting program: C:/Users/Krystal/Desktop/ASM_Practice/fib.exe
[New Thread 16676.0x3a9c (id 2)]

Thread 1 hit Breakpoint 1, 0x00007ff78cf21408 in fib ()
(gdb) continue
Continuing.

Thread 1 hit Breakpoint 1, 0x00007ff78cf21408 in fib ()
(gdb) continue
Continuing.

Thread 1 hit Breakpoint 1, 0x00007ff78cf21408 in fib ()
(gdb) continue
Continuing.

Thread 1 hit Breakpoint 1, 0x00007ff78cf21408 in fib ()
(gdb) info registers rcx rsp rbp
rcx            0x1                 1
rsp            0x15fdb8            0x15fdb8
rbp            0x15fdf0            0x15fdf0
(gdb) ni
0x00007ff78cf21409 in fib ()
(gdb) ni
0x00007ff78cf2140c in fib ()
(gdb) ni
0x00007ff78cf2140d in fib ()
(gdb) info registers rsp rbp
rsp            0x15fda8            0x15fda8
rbp            0x15fdb0            0x15fdb0
(gdb) bt
#0  0x00007ff78cf2140d in fib ()
#1  0x0000000000000002 in ?? ()
#2  0x000000000015fdf0 in ?? ()
#3  0x00007ff78cf21420 in fib ()
#4  0x0000000000000000 in ?? ()
(gdb) set $f1 = $rbp
(gdb) set $f2 = *(unsigned long long*)$f1
(gdb) set $f3 = *(unsigned long long*)$f2
(gdb) set $f4 = *(unsigned long long*)$f3
(gdb) x/gx $f1
0x15fdb0:       0x000000000015fdf0
(gdb) x/gx $f2
0x15fdf0:       0x000000000015fe30
(gdb) x/gx $f3
0x15fe30:       0x000000000015fe70
(gdb) x/gx $f4
0x15fe70:       0x000000000015fea0
(gdb) x/gx $f1+8
0x15fdb8:       0x00007ff78cf21420
(gdb) x/gx $f2+8
0x15fdf8:       0x00007ff78cf21420
(gdb) x/gx $f3+8
0x15fe38:       0x00007ff78cf21420
(gdb) x/gx $f4+8
0x15fe78:       0x00007ff78cf213f2
(gdb) ni
0x00007ff78cf21411 in fib ()
(gdb) info registers rsp rbp
rsp            0x15fd80            0x15fd80
rbp            0x15fdb0            0x15fdb0
(gdb)
```

Задание 6

```
Thread 1 hit Breakpoint 1, 0x00007ff792aa13fd in bad_frame_ret ()
(gdb) info registers rsp rbp
rsp            0x15fe70            0x15fe70
rbp            0x15fe70            0x15fe70
(gdb) x/4gx $rsp
0x15fe70:       0x000000000015fea0      0x00007ff792aa13ed
0x15fe80:       0x00007ff792aa50a8      0x000000000015feb0
(gdb) x/gx $rbp
0x15fe70:       0x000000000015fea0
(gdb) x/gx $rbp+8
0x15fe78:       0x00007ff792aa13ed
(gdb)
```

В ходе эксперимента была создана функция, которая сохраняет три регистра, но восстанавливает только два. С помощью GDB было установлено, что перед выполнением инструкции ret значения RSP и RBP совпадают. При этом по адресу

[RBP] находится сохранённое значение старого RBP, а настоящий адрес возврата расположен по адресу [RBP+8].

Таким образом, из-за неправильного восстановления регистров нарушается баланс стека. Инструкция ret получает неправильный адрес, что может привести к аварийному завершению программы.

Количество операций сохранения и восстановления регистров должно совпадать. Регистры необходимо восстанавливать в обратном порядке.